

A Strong-Field Turn-Off for the Quadrature MOND Kernel: Planetary Safety, a Finite Vacuum Term, and the Offset Reading

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Abstract

The quadrature law $g^2 = g_N^2 + a_0 g_N$, i.e. $\nu(y) = \sqrt{1 + 1/y}$ with $y = g_N/a_0$ (Milgrom 1999, eq. 9), has a slow strong-field tail, $\nu - 1 \simeq 1/(2y)$. Held to all accelerations it gives a constant anomalous acceleration $a_0/2$ in the Solar System, 1279 times the Earth–Mars ephemeris bound, and it makes the vacuum term of the theory’s field Lagrangian (QUMOND, or the nonrelativistic limit of bimetric MOND) diverge linearly. We replace the tail by $\nu_{\text{fix}}(y) = 1 + (\sqrt{1 + 1/y} - 1)/[1 + (y/y_t)^2]$, which equals the law below a turn-off y_t and falls as y^{-3} above it. On SPARC the fixed kernel is indistinguishable from the exact law for $y_t \geq 100$ ($|\Delta\chi^2| \leq 1.06$ on about 3000 points); every planet passes for $y_t \leq 7.7 \times 10^5$ (Mars binds); the vacuum term becomes finite, $\Lambda/a_0^2 \simeq (\pi/4)y_t$. The repair costs one constant. If, in addition, the vacuum energy of the a_0 sector is the observed dark energy (the “offset reading”, an input), y_t is fixed: 128.9 on the footing that defines a_0 from Λ , and 94–103 for measured $a_0 = 1.05\text{--}1.10 \times 10^{-10} \text{ m s}^{-2}$, i.e. the boost turns off at $g_N \approx 10^{-8} \text{ m s}^{-2}$, at the upper edge of present rotation-curve data. The turn-off value depends on its shape ($k = 3, 4$ give 167.5 and 182.4 on the framework footing), so it is not a derivation of the coefficient $\kappa = \frac{1}{2}$, which stays fitted.

1 The problem

The law $g_{\text{obs}}^2 = g_N^2 + a_0 g_N$ fits the radial acceleration relation of SPARC galaxies [LMS16] and is the deep-MOND relation with the next-order term of Milgrom’s 1999 vacuum construction [M99]. Its interpolating function has $\nu - 1 \simeq 1/(2y)$ for $y \gg 1$, the “ $\alpha = 1$ ” tail. Two consequences follow, and both are failures of the law held at all accelerations, not of its galaxy-scale content:

- **Planets.** The anomalous acceleration $(\nu - 1)g_N \rightarrow a_0/2$ is a constant $4.7 \times 10^{-11} \text{ m s}^{-2}$ toward the Sun, 1279 times the Earth bound $3.66 \times 10^{-14} \text{ m s}^{-2}$ and comparably over Mars [SJ06, M09]. Sereno & Jetzer require a tail exponent $\gtrsim 1.5$.
- **Vacuum term.** In QUMOND, or in the $\alpha + \beta = 0$ bimetric theory whose nonrelativistic limit it is [M09b, eq. 3], the field Lagrangian leaves a constant when the field vanishes. With no constant in strong fields this gives $\Lambda/a_0^2 = \frac{1}{2} \int_0^\infty 2y(\nu - 1) dy$ [M09b, eq. 24 with $f(1) = 1$], the same in the exchange-symmetric bimetric class by an exact boundary-term identity (programme lane p25). For $\nu = \sqrt{1 + 1/y}$ the integral grows linearly with any cutoff; cut at the Planck acceleration it gives $\Lambda = \frac{1}{2}a_0 a_P/c^4$, about 10^{59} times the observed value (p27).

2 The turn-off

We keep the law where it is measured and switch the tail off above y_t :

$$\nu_{\text{fix}}(y) = 1 + \frac{\sqrt{1 + 1/y} - 1}{1 + (y/y_t)^k}, \quad k = 2 \text{ (default)}. \quad (1)$$

Below y_t this is the law (deep limit, landmark triplet and sum rule unchanged); above it $\nu - 1 \simeq y_t^k/(2y^{1+k})$. Figure 1 shows the fractional boost for three turn-offs against the planetary bounds converted to $\nu - 1$.

3 Results

Galaxies. With the programme’s SPARC profile likelihood (175 galaxies; Υ fixed at 0.5 with MLS16 cuts, or free per galaxy), the turn-off changes the best χ^2 by +0.07, +0.04, 0.00 (fixed Υ) and -1.06 , -0.67 , -0.01 (Υ free) for $y_t = 100, 128, 10^3$: indistinguishable.

Planets. The largest allowed turn-off is $y_t = 7.7 \times 10^5$ (Mars), 1.77×10^6 (Earth), 5.4×10^7 (Mercury), 1.81×10^8 (Venus). At $y_t = 128$ the worst planet sits at 2.8×10^{-8} of its bound.

Vacuum term. The integral is finite: $\Lambda/a_0^2 = 77.85, 99.81, 784.4$ at $y_t = 100, 128, 10^3$ ($k = 2$). To leading order it is $\frac{1}{2} \int_0^\infty dy/[1 + (y/y_t)^2] = (\pi/4)y_t$; the small- y part of the kernel and a logarithm supply the rest (Fig. 2).

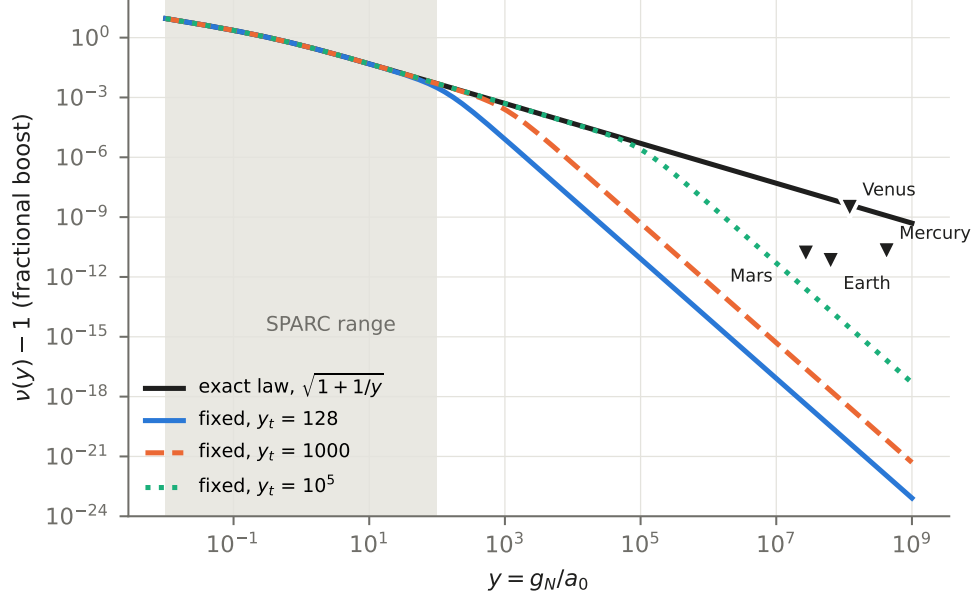


Figure 1: Fractional boost $\nu - 1$ against $y = g_N/a_0$. Black: the exact law. Coloured: the turn-off at $y_t = 128, 10^3, 10^5$ (line style repeats the colour). Triangles: the planetary bounds on $\nu - 1$ (bare, no external-field credit). Shaded: the SPARC range.

4 The offset reading

If the a_0 sector’s vacuum energy is the observed dark energy (an input; programme lane K), the turn-off is no longer free. Then $\Lambda_{\text{obs}}/a_0^2$ fixes y_t :

a_0 (ms ⁻²)	$\Lambda_{\text{obs}}/a_0^2$	y_t ($k = 2$)
9.3603×10^{-11} (defined from Λ , $\kappa = \frac{1}{2}$)	100.53	128.9
1.05×10^{-10} (MeerKAT MIGHTEE-HI)	79.89	102.6
1.0766×10^{-10} (SPARC profile)	75.99	97.6
1.097×10^{-10} (ensemble of nine analyses)	73.19	94.1

On measured a_0 the boost turns off at $y_t \approx 94\text{--}103$, i.e. $g_N \approx 10^{-8} \text{ ms}^{-2}$, just at the upper edge of SPARC (about 5% of points have $y > 10$; the sample tops out near $y \approx 110$). This is the reading’s testable content: rotation-curve, lensing or dynamical data at $10 \lesssim y \lesssim 10^3$ (bulge-dominated inner regions, early-type galaxies) would see the boost fall below the exact law’s by a factor $\approx 1/[1 + (y/y_t)^2]$. It also inherits lane K’s consequence that with $w = -1$ the scale a_0 is constant in time.

What it is not. The turn-off value depends on the shape: $k = 3$ and $k = 4$ need $y_t = 167.5$ and 182.4 on the framework footing. The “round” 128 is 32π divided by the $\pi/4$ of the $k = 2$ integral. The offset reading is a hypothesis, not a consequence of the action. Nothing here derives $\kappa = \frac{1}{2}$ in $a_0 = \kappa c \sqrt{G\rho_\Lambda}$.

5 Summary

A single turn-off removes the quadrature kernel’s two strong-field failures (the planets and the divergent vacuum term) without touching galaxies, at the cost of one constant $y_t \in [100, 7.7 \times 10^5]$. Under the offset reading that constant is set by Λ/a_0^2 and lands at the edge of current data, which makes it testable. The cold mass ($\Omega_c h^2 \approx 0.12$) is still required; no dark-matter particle is added.

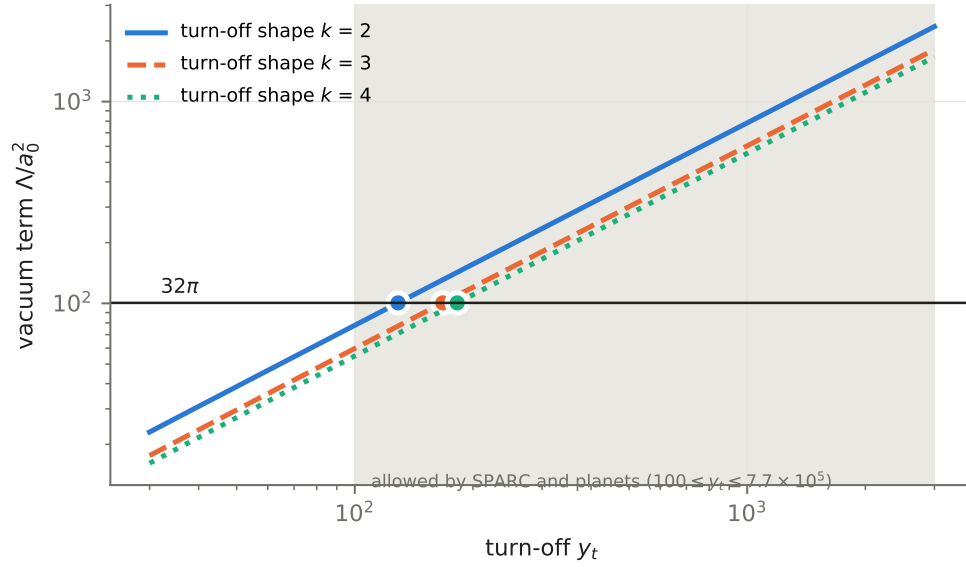


Figure 2: The vacuum term Λ/a_0^2 against the turn-off y_t for three turn-off shapes. Horizontal line: 32π , the framework footing. Dots: where each shape meets it. Shaded: the window allowed by SPARC and the planets.

Reproducibility. `sonnet55_push/puzzle_32pi/p35_kernel_tail_fix.py` (4/4; the exact-law control fails 3), `qwen_claude_field_theory/papers_2026/make_paper41_figures.py`, `PAPER41_figures_numbers.json`, `PAPER41_audit.py`.

References. [LMS16] Lelli, McGaugh & Schombert, AJ 152, 157 (2016). [M99] Milgrom, Phys. Lett. A 253, 273 (1999), astro-ph/9805346. [M09] Milgrom, MNRAS 399, 474 (2009), arXiv:0906.4817. [M09b] Milgrom, Phys. Rev. D 80, 123536 (2009), arXiv:0912.0790. [SJ06] Sereno & Jetzer, MNRAS 371, 626 (2006).